
Unit – 2 : Session – 10 : SLO – 2 | Critical Section Problem — Pseudocode Completion

Answers

TODO 1: flag[1] == true && turn == 1

TODO 2: flag[0] = false;

TODO 3: flag[0] == true && turn == 0

TODO 4: flag[1] = false;

Reason: a process sets its own flag and gives the turn to the other process, then waits only while the other wants to enter and it is the other's turn; on exit it clears its own flag.

Completed pseudocode

```
boolean flag[2] = {false, false};
int turn;

// Process P0
Process0:
    flag[0] = true;
    turn = 1;
    while (flag[1] == true && turn == 1) {           // TODO 1
        // busy wait
    }
    // Critical Section for P0
    print("Process 0 in critical section");
    flag[0] = false;                                 // TODO 2
    print("Process 0 exiting");

// Process P1
Process1:
    flag[1] = true;
    turn = 0;
    while (flag[0] == true && turn == 0) {           // TODO 3
        // busy wait
    }
    // Critical Section for P1
    print("Process 1 in critical section");
    flag[1] = false;                                 // TODO 4
    print("Process 1 exiting");
```